THE EQUIVALENCE RELATION ON FIELD-VALUED POINTS IS TRANSITIVE

THE STACKS PROJECT AUTHORS

Editorial note. Statement span is lines 926-930 of sources/stacks-stacks-properties/pinned-original-source.tex; proof span is lines 932-977, the complete author proof (44 body lines).

Editorial note. Navigation stubs: 1 cited-but-not-included labels are supplied as non-load-bearing pointers with their pinned tags where the snapshot tag index has them.

Editorial note. extra_packages is empty on purpose: the retained preamble already loads xy with options and a second declaration clashes in the assembler wrapper.

Editorial note. Text is retained verbatim, including the author's own typos and placeholders.

Editorial note. Queue index 14 of queue_stacks-properties.json (pb=1465, stmt 926-930, proof 932-977).

Editorial note. stacks-properties wave-1: fresh adjudication accept (2026-10-14); pool scanned while CONTROL.md is down (0 bytes) - §2 row deferred per Chief, claim reported to Chief 08:0x.

Editorial note. Cross-chapter same-label warning (NOT an occupancy conflict, Chief ruling 2026-10-13 evening): label 'lemma-equivalence' also exists in stacks-algebraization, stacks-sites — a different proposition in a different chapter file.

Allowed standard prerequisites (editorial).

- Explicit prerequisite boundary. The cited inputs are prerequisites; the argument in the retained span is not replaced by any cited result.

Source reference. Original lemma 04SS: sources/stacks-stacks-properties/pinned-original-source.tex, source lines 977–977. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Lemma 0.1. *The notion above does indeed define an equivalence relation on morphisms from spectra of fields into the algebraic stack $\mathcal{X}$.*

Proof. It is clear that the relation is reflexive and symmetric. Hence we have to prove that it is transitive. This comes down to the following: Given a diagram

$$\begin{array}{ccccc} \mathrm{Spec}(\Omega) & \xrightarrow{b} & \mathrm{Spec}(L) & \xleftarrow{b'} & \mathrm{Spec}(\Omega') \\ \downarrow{\scriptstyle a} & & \downarrow{\scriptstyle q} & & \downarrow{\scriptstyle a'} \\ \mathrm{Spec}(K) & \xrightarrow{p} & \mathcal{X} & \xleftarrow{p'} & \mathrm{Spec}(K') \end{array}$$

with both squares 2-commutative we have to show that p is equivalent to p'. By the 2-Yoneda lemma (see Algebraic Stacks, Section 04SS) the morphisms p, p', and q are given by objects x, x', and y in the fibre categories of $\mathcal{X}$ over $\mathrm{Spec}(K)$,

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$\mathrm{Spec}(K')$, and $\mathrm{Spec}(L)$. The 2-commutativity of the squares means that there are isomorphisms $\alpha : a^*x \to b^*y$ and $\alpha' : (a')^*x' \to (b')^*y$ in the fibre categories of $\mathcal{X}$ over $\mathrm{Spec}(\Omega)$ and $\mathrm{Spec}(\Omega')$. Choose any field Ω'' and embeddings $\Omega \to \Omega''$ and $\Omega' \to \Omega''$ agreeing on L. Then we can extend the diagram above to

$$\begin{array}{ccccc}
 & & \mathrm{Spec}(\Omega'') & & \\
 & \swarrow^{c} & \downarrow{\scriptstyle q'} & \searrow^{c'} & \\
\mathrm{Spec}(\Omega) & \xrightarrow[b]{} & \mathrm{Spec}(L) & \xleftarrow[b']{} & \mathrm{Spec}(\Omega') \\
\downarrow{\scriptstyle a} & & \downarrow{\scriptstyle q} & & \downarrow{\scriptstyle a'} \\
\mathrm{Spec}(K) & \xrightarrow{p} & \mathcal{X} & \xleftarrow{p'} & \mathrm{Spec}(K')
\end{array}$$

with commutative triangles and

$$(q')^*(\alpha')^{-1} \circ (q')^*\alpha : (a \circ c)^*x \longrightarrow (a' \circ c')^*x'$$

is an isomorphism in the fibre category over $\mathrm{Spec}(\Omega'')$. Hence p is equivalent to p' as desired. $\square$